

ASSESSMENT OF L2 SPANISH PRONUNCIATION ACCURACY VIA AUTOMATIC SPEECH RECOGNITION

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Abstract

This study contributes to the evaluation of non-native Spanish speakers' acoustic production using Artificial Intelligence (AI) tools, specifically Automatic Speech Recognition (ASR) models. In order to determine whether leading ASR models can provide adequate feedback on L2 Spanish pronunciation, we evaluated four models (Wav2Vec, Whisper-large-v2, Whisper-large-v3, and SeamlessM4T) using datasets of non-native Spanish speakers with English, Russian, and German as L1s. Based on a Word Error Rate and Character Error Rate evaluation framework, Whisper-large-v3 and SeamlessM4T demonstrated the highest accuracy for non-native speech recognition. A qualitative and phonetic error analysis revealed that these models struggle when vowel formant boundaries of L2 speakers exceed those of standard Spanish or when voiceless consonants are influenced by phonetic assimilation processes. Additionally, we identified gender bias, with models performing better on female speech than male speech, and substitution errors as the most frequent error type. In conclusion, while ASR models like Whisper-large-v3 and SeamlessM4T perform adequately, an accurate pronunciation assessment for L2 Spanish learners requires their outputs to be complemented by a detailed phonetic analysis.

Keywords: ASR; Whisper; Phonetics; Spanish L2; Non-native speakers



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Avaluació de l'exactitud en la pronunciació del castellà com a L2 mitjançant el reconeixement automàtic de la veu

Resum: Aquest estudi contribueix a l'avaluació de la producció acústica de parlants no nadius de castellà mitjançant eines d'Intel·ligència artificial (IA), concretament models de reconeixement automàtic de veu (ASR). Amb l'objectiu de determinar si els principals models d'ASR poden proporcionar un feedback adequat sobre la pronunciació del castellà com a L2, hem avaluat quatre models (Wav2Vec, Whisper-large-v2, Whisper-large-v3 i SeamlessM4T) utilitzant conjunts de dades de parlants no nadius de castellà, els quals tenen l'anglès, el rus o l'alemany com a llengua materna. Segons els resultats del marc d'avaluació basat en el *Word Error Rate* i el *Character Error Rate*, els models Whisper-large-v3 i SeamlessM4T van demostrar la màxima precisió en el reconeixement de parla no nativa. Una anàlisi qualitativa i fonètica dels errors va revelar que aquests models tenen dificultats quan els límits dels formants vocàlics dels parlants de L2 superen els del castellà estàndard o quan les consonants sordes són afectades per processos d'assimilació fonètica. A més, es va identificar un biaix de gènere, amb un millor rendiment en la parla femenina en comparació amb la masculina, i els errors de substitució com el tipus d'error més comú. En conclusió, tot i que models com Whisper-large-v3 i SeamlessM4T tenen un bon rendiment, l'avaluació precisa de la pronunciació del castellà com a L2 requereix complementar els seus resultats amb una anàlisi fonètica detallada.

Paraules clau: ASR; Whisper; Fonètica; Castellà L2; Parlants no nadius.

Evaluación de la precisión en la pronunciación del español L2 mediante el reconocimiento automático del habla

Resumen: Este estudio contribuye a la evaluación de la producción acústica de hablantes no nativos de español mediante herramientas de Inteligencia Artificial (IA), concretamente modelos de reconocimiento automático del habla (ASR). Con el fin de determinar si los principales modelos ASR pueden proporcionar información adecuada sobre la pronunciación en ELE, evaluamos cuatro modelos (Wav2Vec, Whisper-large-v2, Whisper-large-v3 y SeamlessM4T) utilizando conjuntos de datos de hablantes no nativos de español con el inglés, el ruso o el alemán como L1. Basándose en el marco de evaluación de la tasa de error en palabras y caracteres (WER y CER), Whisper-large-v3 y SeamlessM4T demostraron la mayor precisión en el reconocimiento del habla no nativa. El análisis cualitativo y fonético de los errores reveló que estos modelos tienen dificultades cuando los límites de los formantes vocálicos de los hablantes de la L2 superan los del español estándar o cuando las consonantes sordas están influidas por procesos de asimilación fonética. Además, identificamos un sesgo de género, con un mejor rendimiento de los modelos en el habla femenina que en la masculina, y errores de sustitución como el tipo de error más frecuente. En conclusión, aunque los modelos ASR como Whisper-large-v3 y SeamlessM4T funcionan adecuadamente, la evaluación precisa de la pronunciación para estudiantes de español L2 requiere que sus resultados se complementen con un análisis fonético detallado.

Palabras clave: Reconocimiento de habla automático; Fonética; Español L2; Hablantes no nativos

1. Introduction

Artificial intelligence (AI) technologies have become an influential field of research, which is reflected in numerous applications, including those in second language acquisition (Dodigovic, 2007; Kannan & Munday, 2018; Nianfan, 2021; Almelhes, 2023). Considering the benefit of AI tools in facilitating both learning and teaching tasks, it presents an advantage for second language didactics by providing furnished feedback in the field of L2 acquisition across various subjects: listening (Kim, 2018), speaking (Mohammadzadeh & Sarkhosh, 2018), reading (Kim, 2018; Vidal-Abarca et al., 2014), writing (Hwang et al., 2023), grammar (Rokhayani et al., 2022), and vocabulary (Wang, 2016).

In recent years, Automatic Speech Recognition (ASR), a component of AI, has undergone significant advancements driven by improvements in machine learning techniques (Mehrish et al., 2023). ASR technology enables the conversion of audio signals into text (Yu & Deng, 2016). It is applied across various fields, including human-machine interaction with virtual assistants (Seaborn et al., 2021), transcription services (Malik et al., 2021), and supporting second language acquisition, particularly in phonetics and phonology (Tejedor-García et al., 2021). In the context of developing second language phonetic skills, ASR has been utilized in the creation of speech analysis tools, such as those developed by Hyun (2018) and Anguera et al. (2023), and applied as a functional instrument for assessing second language pronunciation by both L2 educators and learners.

However, as noted by Orihuela Gracia (2021) and San Martín et al. (2024), ASR tools still face challenges in accurately transcribing speech due to the inherent variability of human speech. This is a commonly recognised limitation of ASR systems (Tatman & Kasten, 2017), particularly when handling different accents and variations within a language (Forsberg, 2003). The issue is further amplified in the case of bilingualism, as individuals who speak multiple languages tend to exhibit greater variation in their speech patterns (San Martín et al., 2024). Additional factors, such as gender, age, social dialects, speaking styles, and geographic location, also complicate the performance of these systems (O'Shaughnessy, 2008). Furthermore, ASR tools are primarily trained on standardised databases that mainly consist of native speaker samples, leading them to produce a standard output that may not capture the nuances and irregularities, particularly those produced by L2 speakers. While research into disparities in ASR performance and tools for speech analysis has been extensive in English, studies in other languages, such as Spanish, remain limited (Kantharuban, Vulic, and Korhonen, 2023). Although work by San Martín et al. (2024) explores the evaluation of ASR tools for varieties of Spanish, our study provides insights into the performance of these models on Spanish L2 speech recognition—an area that, to the best of our knowledge, has been under-researched until now. Based on these outcomes, the present paper aims to explore the effectiveness of state-of-the-art ASR systems in analysing pronunciation accuracy among European Spanish L2 speakers with different L1s.

To achieve this objective and in light of the aforementioned outcomes, the present study aims to make the following contributions:

- (1) An evaluation of four currently available ASR models on three distinct datasets of non-native Spanish speakers.
- (2) A qualitative analysis of the output from the ASR models, focusing on the most common types of errors produced.
- (3) A phonetic categorisation of the errors generated by the ASR models.

Moreover, our research aims to contribute to the fields of phonic skills evaluation in Spanish L2 and natural language processing by providing a critical vision of AI tools used for speech recognition. Some of our findings provide further evidence in support of the hypothesis presented

in previously published works. In contrast, others offer novel insights into the specific task of ASR in the context of Spanish speech analysis. Specifically, our results for the best-performing ASR model tested on Spanish speech produced by non-native speakers contrast slightly with previously reported findings on varieties of rural spoken Spanish (San Martín et al., 2024). This and other findings of our study highlight areas that warrant further investigation across additional non-native speech datasets in future research.

The rest of the paper is organised as follows. In Section 2, we present the theoretical and methodological background related to our work. Subsequently, we describe the data recompilation and experimental design of our work. In Section 4, we present the results obtained with the ASR systems applied to analyze our corpus. Section 5 is dedicated to conclusions and a discussion of the advantages and disadvantages of using ASR tools for analyzing the pronunciation accuracy of Spanish L2 speakers, based on the qualitative analysis of the results presented in Section 4.

2. Related work

Pronunciation acquisition has traditionally been viewed as a mechanical and unrealistic process, primarily focusing on the isolated production (through repetition and memorization) of segmental sounds (phonemes) in a decontextualised manner (Cantero, 2020). This approach assumes that pronunciation is a fixed, standard phenomenon, as though there is a single phonological system and, therefore, one "correct" way of speaking that the learner should aim to achieve. However, Cantero (2020) posits that pronunciation should be understood as encompassing the entire phonetic dimension of language. This view is further supported by Brum Barreto, Corrêa Lopes, and Fonseca de Oliveira (2024), who aim to develop an integrated didactical approach to phonetic component acquisition for Spanish L2 learners. According to all aforementioned authors, pronunciation is a complex and multidimensional aspect of communicative competence, which requires a similarly complex approach for its assessment. Based on these insights, this section aims to explore existing ASR models and examine their potential for evaluating pronunciation accuracy, to determine whether these models adequately account for the complex nature of pronunciation.

2.1. ASR models

As noted by San Martín et al. (2024), most ASR models today are typically trained on either English-only data or multilingual data, making the use of multilingual models mandatory for speech recognition in Spanish. Among the available open-source alternatives placed on the HuggingFace open-access repository, models such as Conformer-CTC (Gulati et al., 2020), PocketSphinx (Huggins-Daines et al., 2006), and Wav2Vec (Baevski et al., 2020) are commonly used.

However, the state-of-the-art ASR multilingual models are based on the Whisper architecture, as indicated by the ASR leader board in November 2024 (Srivastav et al., 2023). Previous studies that analysed L2 speakers' pronunciation accuracy using ASR models, such as those by Graham and Roll (2024), employed Whisper along with SeamlessM4T (Lin et al., 2024) and Wav2Vec (Shibano et al., 2021). While these studies demonstrated promising results, their evaluations were limited exclusively to English. To provide a consistent contribution to the field of pronunciation accuracy using ASR in Spanish, we used the cited models previously tested on English ASR task, as well as the models previously reported by San Martín et al. (2024) evaluated on Spanish peninsular varieties by San Martín et al. (2024), which are the following: Wav2Vec¹ by Baevski

¹ We use multilingual Spanish version for our experiments:

<https://huggingface.co/jonatasgrosman/wav2vec2-large-xlsr-53-spanish>

et al. (2020), Whisper by Radford et al., (2023) and SeamlessM4T by Barrault et al. (2023). Specifically, we used Wav2Vec (Baeovski et al., 2020), Whisper (Radford et al., 2023), and SeamlessM4T (Barrault et al., 2023). These models were selected for two primary reasons: the need for further validation of their performance on Spanish, particularly beyond varieties that Spanish L2 speakers may provide, and their accessibility, as all three models are openly available on the HuggingFace repository.

All the models selected for this study are available in various sizes (base, small, medium, and large). However, we opted to use the large versions for all models, as they demonstrated the best performance according to the HuggingFace sources (Srivastav et al., 2023), where the models are stored. In the case of the Whisper ASR model, which has three versions in large size, we included two distinct versions due to the following reasons: the large-v2 model was trained for a longer duration of data compared to the large version. In contrast, the large-v3 model was trained using additional data, part of which was automatically generated with the large-v2 model. Although San Martín et al. (2024) observed minimal differences between the large-v2 and large-v3 versions of Whisper, our study revealed slight variations, particularly in the qualitative analysis of output data. Consequently, the final selection of models for this research consists of four models: Wav2Vec, Whisper-large-v2, Whisper-large-v3, and SeamlessM4T.

Regarding related works that applied ASR approaches to address the challenge of non-native speech recognition, studies by Vu et al. (2014) and Radzikowski et al. (2021) proposed modifying the input data of non-native English speakers by transforming their phonic features into those of native speakers. Specifically, their approach involved a two-step system: first, the system converts a sound wave file into a spectrogram using an existing ASR model architecture. Then, it modifies the L2 phonetic features by automatically comparing the spectrograms of non-native speakers with those of the same utterances spoken by native speakers, ultimately converting the non-native spectrogram into a modified, native-like version.

This approach, however, contradicts the holistic perspective of L2 pronunciation acquisition proposed by Cantero (2020) and Cantero & Font-Rotchés (2025), which criticizes the idea of a single "correct" way of speaking as the highest standard for learners to achieve. Instead, the holistic view acknowledges the diversity of pronunciations as an integral part of language learning. Additionally, training ASR systems to erase L2 speakers' phonetic features effectively introduces biases toward native-like speech. The aim of ASR models should not be to convert L2 speech into native speech but rather to accurately detect and analyse L2 phonetic patterns within the non-natives' speech.

This latest approach is further developed in the work by Shibano et al. (2021), who introduced the Wav2Vec ASR model. The authors trained and evaluated the model using the raw phonic features of non-native English speakers without modifying them. Wav2Vec processes raw audio input, directly extracting acoustic patterns from the waveform. It learns to identify or predict functional speech elements such as phonemes, intonation, and pauses through data-driven, statistical learning. This method enables the model to develop robust, context-aware word representations without relying on precomputed spectrograms. The evaluation results of this model outperform those of models that modify non-native phonetic patterns, suggesting that individual speaker differences may significantly influence the extent to which ASR model performance can improve.

As for the Whisper ASR model, as described by Radford et al. (2023), it transforms speech signals into spectrograms, a frequency-time representation that captures acoustic features of a speech contour. This process filters out certain raw acoustic features, such as noise and pitch variations, which may limit its ability to capture detailed phonetic-phonological characteristics of a speaker when compared to models like Wav2Vec, which process raw waveforms. At the same time, Whisper relies on structured input data and learning to group acoustic features into recognizable prosodic constituents (Nespor & Vogel, 1986), ranging from phonemes to syllables, phonological

words, and phonological phrases. The model processes data by learning contextual relationships, such as how phonemes combine into prosodic constituents, and subsequently into morphemes and words.

Finally, SeamlessM4T, developed by Barrault et al. (2023), also uses spectrograms but is specifically designed for multilingual speech processing. This model is trained to capture language-specific phonetic features, enabling it to recognise intricate acoustic patterns across languages. As a result, SeamlessM4T is sensitive to linguistic variations, including segmental features (e.g., nasalization and palatalization) and suprasegmental features (e.g., stress and intonation) related to these variations. Given its focus on multilingual tasks, SeamlessM4T must account for prosody and rhythm in a language-agnostic manner. This capability is crucial for applications such as transcription, where the phonetic and prosodic structure of one language must be effectively mapped onto another.

To summarize, each ASR model has its advantages and limitations. In this context, Wav2Vec processes raw audio data without excluding any of its properties, thereby retaining the full spectrum of acoustic features. However, its performance on recognition tasks depends on the quality and diversity of the training data. Since the model learns from this data and makes statistical predictions (even when leveraging neural networks), inadequate representation of a language's segmental and suprasegmental variations during pretraining may result in suboptimal performance when assessing the pronunciation accuracy of L2 speakers. In contrast, the Whisper model relies on context-related data. However, its filtering of certain acoustic features during input processing may lead to the omission of L2-specific segmental and suprasegmental elements, which are critical for accurate pronunciation assessment. Lastly, the SeamlessM4T model is trained on multilingual datasets, which provides it with a broad understanding of the acoustic features of various languages. However, its behaviour in complex contexts, such as recognizing and evaluating L2 speech, remains uncertain. There is limited insight into how effectively it handles scenarios where acoustic features from multiple languages overlap.

Given these insights, our research aims to identify which ASR model performs best in recognizing and assessing L2 Spanish speech pronunciation. Additionally, we aim to analyze the types of errors these models produce in this context, as outlined in Section 1. To ensure a robust contribution to these topics, we outline the methodology used to evaluate the performance of these ASR models in the following subsection.

2.2. Evaluation methodology

2.2.1. ASR assessment metrics

To evaluate the performance of ASR models in recognizing the speech of non-native L2 Spanish speakers, we employed the *Word Error Rate* (WER) metric (Woodard & Nelson, 1982), a widely used standard for assessing the accuracy of ASR systems. It is calculated by comparing a reference sentence with an automatically generated sentence using the following formula:

$$W E R = \frac{S + D + I}{N}$$

In this case, S represents the number of substitutions, D the number of deletions, I the number of insertions, and N the total number of words in the reference sentence. A lower WER value indicates better performance. Notably, before computing the WER, both the reference and automatically generated sentences are normalised by removing punctuation marks and converting all text to lowercase. This normalization is essential since punctuation is an orthographic feature of written language, rather than an oral transcription, making it logical to evaluate the text in its normalized form.

Additionally, we employ the Character Error Rate (CER), as proposed by Morris, Maier, and Green (2004), which is a metric similar to WER but operates at the character level instead of the word level. CER is calculated using the following formula:

$$CER = \frac{S_c + D_c + I_c}{N_c}$$

In this formula, S_c stands for the number of character substitutions, D_c denotes the number of character deletions, I_c indicates the number of character insertions, and N_c is the total characters in the reference sentence.

According to the aforementioned authors, CER is particularly effective for comparing models in tasks such as Automatic Speech Recognition, especially when dealing with multilingual datasets where WER may not be a sufficient indicator due to language diversity. In our study, we analyse a corpus of L2 Spanish speakers with various L1 backgrounds. Although our dataset does not consist of multilingual samples, it may still exhibit complex multilingual acoustic features due to L1 interference in the pronunciation of L2 Spanish speakers. Subsection 3.2 provides a clearer example of how these formulas work.

As noticed by Morris, Maier, and Green (2004), both WER and CER metrics may not provide all the insights into the specific nature of speech recognition errors, requiring further investigation to identify the primary sources of these failures. Therefore, there is an emerging necessity to incorporate phonetic analysis to ensure a comprehensive and holistic evaluation of the ASR models' output.

2.2.2. Phonetic assessment of ASR results

As explained in Section 2.1, the selected ASR models process acoustic input using different approaches. Some models operate directly on raw audio data, while others first convert the input into spectrograms. These differing processing methods pose challenges for conducting both segmental and suprasegmental analyses of the ASR results, as not all models produce spectrograms during their processing stages or draw pitch contours. However, the standard output across all models is an automatically generated sentence, which can be assessed from the perspective of phonetic transcription accuracy. In other words, we can evaluate the phonetic accuracy of the transcriptions generated by these ASR models for L2 Spanish non-native speakers. Consequently, if the model produces an output for the processed acoustic input as “todas las mañanas *eren* iguales,” and we compare it to the reference utterance “todas las mañanas *eran* iguales,” an error in the transcribed text is observed. This allows us to perform a phonetic analysis specifically on the erroneous segment.

In order to provide the assessment of the L2 Spanish learners' phonetic accuracy, we rely on the Revised Speech Learning Model (SLM-r) by Flege and Bohn (2021) builds upon the original Speech Learning Model (SLM) proposed by Flege (1995), and sustained in the work Cao, Font-Rotchés, & Rius-Escudé (2024), which suggests that first language (L1) and second language (L2) sound categories coexist within the same phonological dimension. This interaction opens the possibility for mutual influence between the two languages, mediated by processes of assimilation and dissimilation. When L2 sounds differ from L1 sounds, learners are more likely to form new phonetic categories after receiving sufficient exposure to these distinct L2 sounds. This phenomenon is known as dissimilation. Conversely, sounds that are similar across L1 and L2 present a greater challenge for learners, as these corresponding sounds often merge into a single category. As a result, the acoustic properties of the sounds in both languages tend to influence one another in a process known as assimilation (Flege, 2005, 2007, 2011).

In line with this methodological approach, we provide a robust qualitative phonetic analysis of the output of ASR models by extracting the values of the first two formants (F1 and F2) of vowel

sounds that the model transcribes incorrectly using the Praat acoustic analysis program (Boersma & Weenink, 2018). For the TextGrid annotations, we utilized the Schiel (1999) and Kisler et al. (2017) software tools to automate the generation of TextGrids. F1 and F2 intensity values were taken as reference at the midpoint of each sound, since they are considered fundamental parameters for the characterization of manner and place of articulation² (Quilis & Esgueva, 1983; Bradlow, 1995). For consonants, our acoustic analysis is based on their distinctive features (Quilis, 1981; Demestre et al., 2006) and acoustic measurements such as VOT (Voice Onset Time) and intensity (Flege & Eefting, 1987; Hualde, 2005; Gil & Llisterri, 2024). For both vowel and consonant phonetic analyses, we differentiate the sounds produced by male and female speakers.

Regarding the processing of the extracted sound values³, we rely on the methodology proposed by Cao, Font-Rotchés, and Rius-Escudé (2024), as this approach has demonstrated promising results and has been statistically validated for analysing sounds produced by non-native L2 Spanish speakers. This quantitative analysis is intended to validate the qualitative evaluation of the ASR results. According to the aforementioned authors, during statistical data processing, the results of non-native L2 Spanish speakers are compared with those of native speakers, taking into account their *precision*⁴. In the cited study, precision is defined as the degree of agreement between measured values when the same quantity is measured repeatedly. Using the precision parameter, it is possible to observe the stability and dispersion of manner and place of articulation during sound production. Cao, Font-Rotchés, and Rius-Escudé (2024) compared the centrality (or dispersion) of sound articulation between native Peninsular Spanish speakers and non-native Spanish speakers with Chinese L1, following the formula proposed by Wen et al. (2016):
$$p1(p2) = \frac{F1(F2)Max.L2 - F1(F2)Min.L2}{F1(F2)Max.L1 - F1(F2)Min.L1}$$

$$p1(p2) = \frac{F1(F2)Max.L2 - F1(F2)Min.L2}{F1(F2)Max.L1 - F1(F2)Min.L1}$$

The correlation between the degree of dispersion of non-native Speaker sound (L2) and the degree of dispersion of a Spanish standard (L1) sound, in either the F1 or F2 dimension, is calculated by multiplying the coefficient of the F1 dimension (p1) by the coefficient of the F2 dimension (p2), using the following formula:

$$p = p1 \times p2$$

The results indicate that if the *p*-value is less than 1, the learners' pronunciation of a sound is more stable or more approximate to native speakers.

After establishing the methodological and theoretical framework for this study, we outline the procedure for compiling the dataset used to evaluate the selected ASR models, as well as the process of computer-assisted analysis of these corpora.

² Although this is the approach we use for vowel analysis in the present study, in future work we aim to incorporate intensity points, as suggested by the recent study by Cantero (2019).

³ This approach is also applicable in cases where formant transitions occur within a syllable. During the coarticulation process, the place and manner of articulation of one sound may be influenced by an adjacent sound (Demestre, Llisterri, Riera, & Soler, 2006).

⁴ In the original study by Cao, Font-Rotchés, and Rius-Escudé (2024), two key aspects—accuracy and precision—were considered for evaluating vowel production by non-native Spanish speakers. However, in our work, we do not include accuracy measurements, as they are specifically used for analysing interactions between multiple factors and performing pairwise comparisons, which are not relevant to our analysis.

3. Experimental design

3.1. Corpus

For this research, we first processed two existing corpora of non-native English speakers using four selected ASR models. These corpora were used for a pilot test of the models to assess their overall performance, which is discussed in Section 4.

One of the datasets used is the AMI Meeting Corpus (Swietojanski et al., 2013), which consists of 100 hours of meeting recordings. For our experiment, we selected 300 sample utterances from this corpus. These recordings were conducted in specially equipped, instrumented meeting rooms across Edinburgh. The corpus comprises two types of meetings: scenario-based meetings, in which four participants assume predetermined roles (e.g., project manager, designer), and non-scenario-based meetings, which are natural and spontaneous discussions on a range of topics. Scenario-based meetings make up approximately 70% of the corpus. Each meeting typically involves four participants, with discussions conducted in English, mostly among non-native speakers. The acoustic signals are captured using specialised professional equipment, including individual head microphones. This corpus does not specify the L1 background of non-native English speakers. Nonetheless, this aspect is considered valuable, as it enables the evaluation of ASR model performance in recognizing English speech across diverse acoustic inputs.

The second pilot dataset is the L2-ARCTIC corpus, created by Zhao et al. (2018). It is a speech dataset of non-native English, designed for research in voice conversion, accent conversion, and mispronunciation detection. This initial release features recordings from ten non-native English speakers, whose first languages (L1s) are Hindi, Korean, Mandarin, Spanish, and Arabic. Each L1 group contains recordings from one male and one female speaker. Each speaker recorded approximately one hour of read speech. The corpus also provides phonetic transcriptions in Praat's TextGrid format. For our preliminary experiments, we utilised 300 examples from recordings of L1 Spanish speakers learning English as a second language. This approach aimed to evaluate the models' performance regarding the specific acoustic features of this L2 scenario, where the phonetic and phonological characteristics of L1 Spanish interact with those of L2 English.

Both of these datasets contributed to our study by providing preliminary quantitative data on the performance of the four ASR models in recognizing non-native speech. None of these corpora was used in the training of the four ASR models, ensuring the validity of the evaluation experiments.

Our L2 non-native Spanish speakers' corpus consists of three groups based on their L1 language. The first group includes English L1 learners of Spanish. The informant group comprises twenty speakers (10 females and 10 males), aged 17-22. The audio files, which are approximately 4-7 seconds in duration, were compiled from the Spanish Learner Language Oral Corpora (SPLLOC) (Mitchell et al., 2008-2010). This corpus includes recordings of British students with a B2-C1 level of Spanish, as reported by the authors. These students are native English speakers who have learned Spanish as a second language within the UK's educational curriculum. It is worth noticing that bilingual individuals and those with extensive social ties to Spanish speakers were excluded from the corpus. Spanish proficiency levels were assessed based on the number of hours British students had spent studying Spanish as a second language. According to the corpus authors, students aged 17-18 who have completed at least 750 hours of Spanish instruction are classified at the B2 level of the CEFR (Council of Europe, 2020). In contrast, students aged 21-22 who have reached the C1 level have completed more than 895 hours of Spanish language education and have had one year of experience living in Spanish-speaking countries. This corpus was previously

used by Sarymsakova & Martín Rodilla (2024) to assess the accuracy of software for automatic analysis of L2 Spanish intonation, yielding promising results. Therefore, we consider it eligible for our experiments in this study.

The second dataset of L2 Spanish learners includes samples from the work of Sarymsakova and Martín-Rodilla (2023), which consists of 5 female learners (aged 24-34) and four male learners (aged 18-35), all of whom have Russian as their L1. According to the authors, all participants are at the B1-B2 proficiency level and do not currently reside in a Spanish-speaking country. In total, 36 audio recordings were made, from which we selected 20 recordings, each ranging from 4 to 7 seconds in duration, matching the duration of the English-speaking L2 learners' corpus. These recordings were part of a didactic exercise in which learners repeated utterances produced by native speakers to improve their L2 Spanish pronunciation. As such, the speech is classified as controlled and non-spontaneous.

Finally, we compiled a dataset of German-speaking L2 Spanish learners, consisting of 10 female and 10 male-produced samples, matching the duration of both the Russian and English-speaking L2 learners' corpora. These samples were sourced from Creative Commons-licensed, freely distributed videos on YouTube⁵. Specifically, the recordings are from DELE examinations (*Diploma de Español como Lengua Extranjera*) at B1-B2 levels. Due to the limited availability of data with a free distribution license, we compiled these samples from a single male and a single female speaker. All samples were taken from the first part of the oral competence examination, which involves spontaneous interaction between the examiner and the candidate. Additionally, we compiled 10 samples from a Fono.ele⁶ source, as experiments with the previous female samples showed extremely low metrics (reported in Section 4). We compiled samples from a female speaker with a B1 level of Spanish proficiency, aged 18-25, as reported by the source author (Canales, 2011). These samples were produced during brief, spontaneous conversations with an interviewer.

In summary, the samples from all three datasets share the same duration (4-7 seconds) and consist of speakers with different L1 backgrounds learning Spanish as a second language at proficiency levels ranging from B1 to C1 (from intermediate to advanced). Each dataset maintains a balanced gender distribution, with 10 male and 10 female speakers per corpus. Although it was not possible to collect samples from multiple speakers for the German dataset, as was done for the other datasets, we ensured that the utterances differed in terms of lexical and grammatical content. All these aspects are expected to contribute to the validation of experiments with the selected ASR models, which are described in the following subsection.

3.2. Experiments

We developed two categories of Python scripts. The first category conducts experiments with Automatic Speech Recognition models and computes the WER and CER metrics. The second category performs qualitative error analysis based on key components of WER and CER by calculating the insertions, deletions, and substitutions for each processed acoustic input.

The script responsible for conducting experiments operates as follows: it requires a WAV audio file (in this case, produced by non-native speakers) and a corresponding transcription in a text file as a reference. The script's algorithm preprocesses the reference transcription by removing punctuation and converting all text to lowercase, facilitating the subsequent calculation of the WER and CER metrics, as described in Subsection 2.2.1. The script then opens a text file

⁵ Male DELE candidate video: <https://youtu.be/14fpP9bBmuo?si=5s4x62xCwWNDYcjP>

Female DELE candidate video: <https://www.youtube.com/watch?v=oEF38OiMp-E>

⁶ <https://fonoele.web.uah.es/corpus-muestra.php>

containing paths to all the audio files it needs to process and stores them in a list of audio paths. Similarly, it opens another text file containing paths to the corresponding reference transcription files and stores them in a list of transcript paths. The script iterates over each pair of audio and transcript files (it reads the audio data and its corresponding transcript).

Subsequently, the script employs one of four ASR models—Wav2Vec, Whisper-large-v2, Whisper-large-v3, or SeamlessM4T—from Hugging Face's Transformers library to process the audio input. The selected ASR model processes the audio and generates a predicted utterance.

Finally, the Character Error Rate and Word Error Rate metrics are computed using the formulas provided in Subsection 2.2.1. The algorithm compares the ASR output with the reference transcription to determine the metrics. For each pair of predicted and reference utterances, the script outputs the WER and CER values, providing a detailed assessment of the ASR model's performance, including how accurately the model recognised the speech and how closely the recognition aligns with the original reference.

Regarding the second category of scripts, it serves as a complement to the previously described script by providing a detailed analysis of each type of error—insertions, deletions, and substitutions—through a comparison of the reference utterance and the generated ASR output. Below is an example illustrating its functionality:

Reference: con eso entiendo que es usted muy listo para dar las explicaciones

Predicted: con eso si entiendo que eso esté muy listo para dar [las] explicaciones

In this case, insertions are highlighted in green, substitutions in red, and deletions in violet. Using the formula provided in Section 2.2.1, the WER can be calculated as follows:

$$W E R = \frac{S(2) + D(1) + I(1)}{N(12)} = 0.33$$

This indicates that the error rate in this case is 33%, assuming the reference sentence represents 100%.

$$C E R = \frac{Sc(7) + Dc(3) + Ic(2)}{Nc(55)} = 0.21$$

For the CER metric, we calculate it by taking the total number of characters in the reference (N_c) as 55 (excluding spaces). With seven insertions, three deletions, and two substitutions, we obtain a character error rate (CER) of 21%.

Taking into account the experimental design described in the present section, we proceed to present the results of the experiments in the following section.

4. Results: L2 Spanish Speech Recognition Task

Hereafter, we present the results of evaluating four ASR models (Wav2Vec, Whisper-large-v2, Whisper-large-v3, and SeamlessM4T) on two pilot datasets of non-native English speakers (AMI Meeting and L2-ARCTIC corpora) and three datasets comprising non-native Spanish speakers (L2). These three datasets were compiled using samples produced by speakers with different L1s: English, Russian, and German.

As described in Section 3.1, for our pilot experiments, we selected two corpora containing recordings of non-native English speakers with different L1 acoustic inputs. This selection was performed to evaluate the performance of the ASR models in the speech recognition task, considering the influence of these phonetic and phonological characteristics. The results of these preliminary tests are presented in Table 1.

Dataset	Wav2Vec		Whisper-large-v2		Whisper-large-v3		SeamlessM4T	
	WER%	CER%	WER%	CER%	WER%	CER%	WER%	CER%
AMI Meeting corpus	53	33	31	20	26	15	54	46
L2-ARCTIC corpus	23	10	18	10	18	10	15	8

Table 1. The average WER and CER metrics of the four ASR models evaluated on 300 examples derived from the AMI and L2-ARCTIC datasets.

As presented in the table above, the Whisper-large-v3 ASR model achieved the lowest WER and CER among the evaluated models. Specifically, when tested on the AMI corpus, it achieved a WER of 26% and a CER of 15%. On the L2-ARCTIC corpus, these metrics improved further, with a WER of 18% and a CER of 10%. Additionally, Whisper-large-v2 and Whisper-large-v3 demonstrated identical WER and CER metrics when evaluated on samples from English learners with L1 Spanish backgrounds. These findings align with the conclusions of San Martín et al. (2024), which suggest that the large versions of Whisper exhibit no significant differences in their performance.

Regarding the performance of four speech recognition models across three datasets of L2 non-native Spanish speakers, we present the evaluation results in Table 2.

Dataset	Wav2Vec		Whisper-large-v2		Whisper-large-v3		SeamlessM4T	
	WER%	CER%	WER%	CER%	WER%	CER%	WER%	CER%
English L1	30	9	16	6	12	4	11	5
Russian L1	14	5	23	10	12	8	17	7
German L1	16	5	40	25	18	15	19	10
Average	20	6,3	26,3	13,67	14	9	15,67	7,33

Table 2. The WER and CER percentages of the four ASR models evaluated on three L2 Spanish speakers corpora with L1 acoustic features from English, Russian, and German.

Based on the data presented above, the overall performance of the ASR models can be summarised as follows. On average, Whisper-large-v3 exhibits the best performance, with the lowest WER of 14% and the second-lowest CER of 9%. SeamlessM4T follows closely, with the second-lowest average WER of 15,67% and the CER of 7,33%. Wav2Vec yields more varied results, achieving an average CER of 6.3% but a higher average WER of 20%. Within these observations, it becomes evident that Whisper-large-v2 generally underperforms compared to the other models, which contradicts the findings of San Martín et al. (2024), who suggested a similar performance across the large versions of Whisper. Our results indicate that Whisper-large-v3 significantly outperforms its counterpart, Whisper-large-v2.

Regarding performance on L1-specific Spanish-speaking datasets, we observe the following trends: for English-native Spanish learners, SeamlessM4T performs best with 11% WER and 5% CER, closely followed by Whisper-large-v3, which achieves 12% WER and 4% CER. For Russian-native speakers, the Wav2Vec model outperforms others with 14% WER and 5% CER,

with Whisper-large-v3 ranking as a close second. Lastly, for German-native speakers learning Spanish as a second language, the Wav2Vec model demonstrates the best performance with 16% WER and 5% CER, followed by SeamlessM4T and Whisper-large-v3.

In general, the ASR model performance trends can be characterised as follows: Whisper-large-v3 shows the most consistent performance across languages, maintaining balanced low WER and CER metrics, which aligns with the evaluation results on two pilot datasets (AMI and L2-ARCTIC corpora). SeamlessM4T also shows strong adaptability across languages, performing well in all three non-native Spanish-speaking datasets. Notably, the improvement from Whisper-large-v2 to Whisper-large-v3 is significant. This finding represents a new contribution, differing from the conclusions of San Martín et al. (2024).

Relaying on these findings, we conducted a quantitative analysis of the distribution of WER and CER metrics across the selected ASR models, categorized by the speakers' gender, as shown in Figure 1.

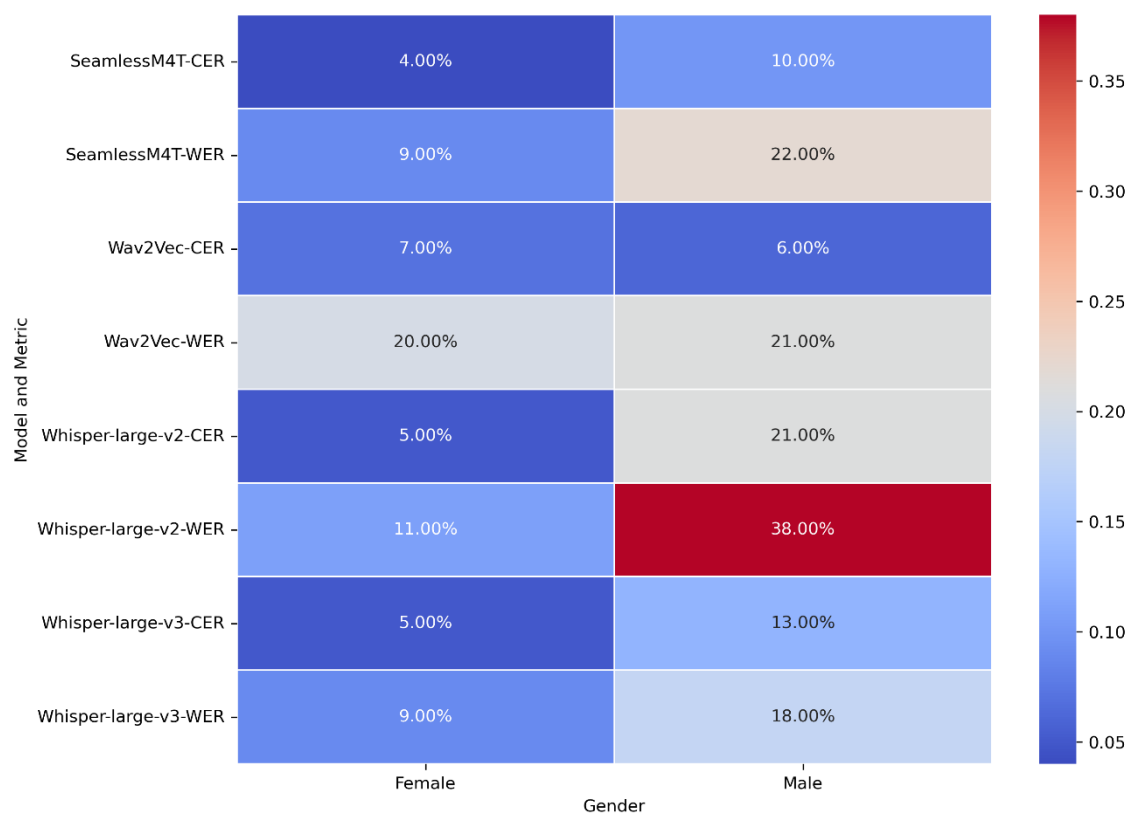


Figure 1. Error rate distribution according to speaker gender across four ASR models.

The heatmap reveals a consistent trend across all four ASR models and both WER and CER metrics, showing higher error rates for male speakers compared to female speakers. This disparity is especially evident in the Whisper-large-v2 model, which exhibits the most significant difference in error rates: 38% WER and 21% CER for male speakers versus 11% WER and 5% CER for female speakers. In contrast, Wav2Vec displays the most consistent performance across genders, with relatively small differences in error rates for both male and female speakers (WER ranging from 20% to 21% and CER from 6% to 7%). As for Whisper-large-v3 and SeamlessM4T, these models demonstrate improved performance over Whisper-large-v2, particularly for male speakers.

When comparing the two metrics, CER consistently shows lower error rates than WER across all models and speaker genders, with the gender gap being smaller for CER than for WER. For

female speakers, SeamlessM4T and Whisper-large-v3 exhibit the lowest error rates, while for male speakers, Wav2Vec and Whisper-large-v3 outperform the other models.

In summary, the aforementioned findings reveal a systematic bias in speech recognition performance, with male speakers generally exhibiting higher error rates. This suggests the need for more balanced training data, considering speaker gender, to fine-tune these models and address this gender disparity. The transition from Whisper-large-v2 to Whisper-large-v3 exhibits significant improvements, particularly in reducing the gender gap and lowering overall error rates, which aligns with previously presented data (Tables 1 and 2). Wav2Vec maintains the most consistent performance across genders, although it does not present the best accuracy. Models like SeamlessM4T and Whisper-large-v3 offer better overall performance but with slightly more gender variation. Finally, the consistently lower CER compared to WER indicates that these models are more effective at recognizing individual speech segments (such as phonemes or syllables) for L2 Spanish speakers, associating them with individual characters. However, they still struggle with constructing more complex speech units, such as utterances, likely due to the interference of L2 acoustic features that the models have not been trained to recognise.

5. Discussion: Qualitative and phonetic analysis

To gain a better understanding of the factors contributing to the superior performance of specific ASR models in L2 Spanish speech recognition, this section presents the results of a qualitative and phonetic analysis, focusing on the identification of the categories of the most frequent errors and examining their acoustic characteristics to provide insight into the underlying causes. Following this line, a quantitative analysis was conducted on three types of errors that compose the WER and CER metrics. The results of this analysis are presented in Figure 2 (see below).

According to the data presented in Figure 2, substitution errors are the most frequent type of error across all models and genders, followed by deletions and insertions. Deletions and insertions occur less, with deletions being slightly more common than insertions in most cases. Male speakers generally exhibit higher error rates than female speakers across all models and error types, corroborating the findings previously reported in Figure 1. This discrepancy is particularly notable for the Whisper-large-v2 model, where male speakers demonstrate significantly higher substitution error rates. Conversely, the SeamlessM4T and Whisper-large-v3 models display the lowest overall error rates for both genders, whereas Wav2Vec records the highest error rates, particularly for substitution errors. This analysis also confirms a degree of gender bias, as observed earlier in the Figure 1 analysis, with models performing better on female speech than on male speech.

We also identified a less common type of error, specifically hypercorrection, within the Whisper-large-v3 model, as demonstrated in the following example:

Reference: países nuevos que no conozco culturas nuevos

Whisper-large-v3: países nuevos que no conozco o culturas nuevas

This example provides valuable insights into L2 Spanish pronunciation assessment, as the model offers a correction via an insertion and a substitution to the non-native speaker's utterance. In doing so, it highlights a segment that may require improvement, effectively providing feedback that could lead to pronunciation refinement.

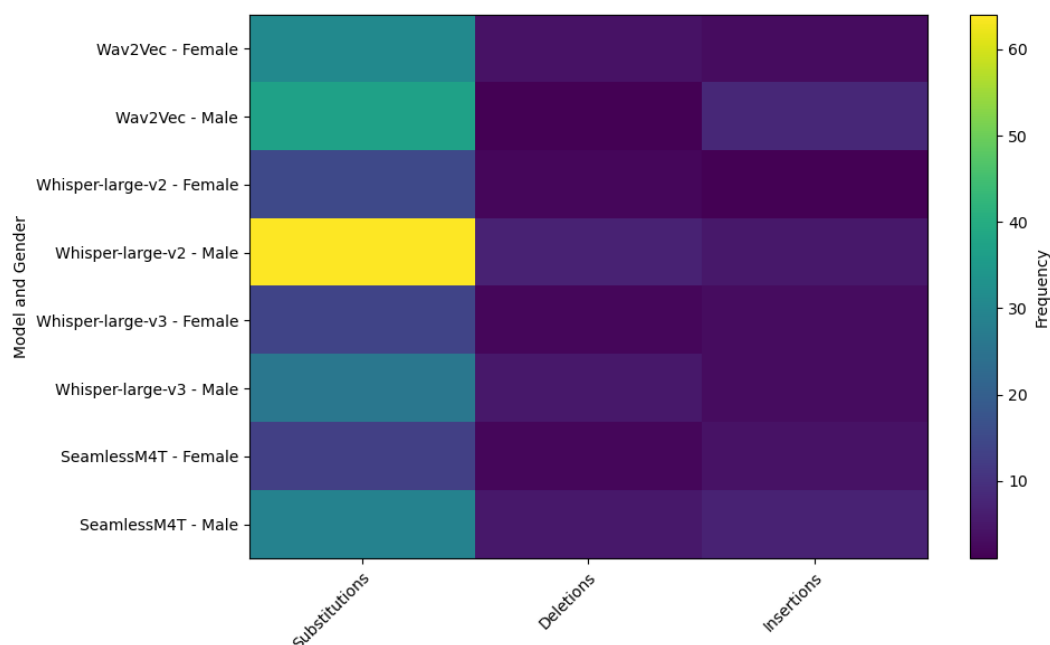


Figure 2. Substitutions, deletions, and insertions distribution across ASR models and datasets.

Based on these outcomes and the results previously reported in subsection 4.1, we observe that the best-performing models for speech recognition of non-native L2 Spanish speakers are Whisper-large-v3 and SeamlessM4T. These models achieve the lowest WER and CER metrics, as well as the lowest frequencies of substitutions, deletions, and insertions across all three datasets. However, this trend requires further validation through a phonetic analysis, based on the outputs produced by these ASR models. To facilitate this analysis, we compiled the following examples from the predicted utterances of Whisper-large-v3 and SeamlessM4T:

Reference: mientras javier bebía agua silvia fregaba los platos

*Whisper-large-v3: mientras **javi** bebía agua silvia fregaba los **plazos***

*SeamlessM4T: mientras javier bebía agua silvia fregaba los **brazos***

These examples were produced by female native English speakers learning Spanish as a second language (L2). For the Whisper-large-v3 model, two substitution errors were identified, whereas the SeamlessM4T model exhibited only one error. To explain these substitution errors, we analyzed the reference audio input.

According to the results from the Whisper-large-v3 model, there is a reduction in the toneme (the last accented syllable) in the acoustic realization of the nominal syntagma “Javier,” which is characterised by the presence of a diphthong. As noted by Demestre, Llisterri, Riera, and Soler (2006), diphthongs are distinguished by a continuous movement of the formants from the proper position of the first vowel of the diphthong to the characteristic position of the second vowel. However, in the case of female native English speakers learning Spanish as an L2, the positions of F1 and F2 show minimal change, with no specific distribution in the first two formants' positions, as illustrated in Figure 3. To analyse the diphthong's acoustic realization, we extracted the F1 and F2 values for both vowels using Praat software: for first (F1 (L2) max = 508 Hz; F1 (L2) min = 382 Hz; F2 (L2) max = 2022 Hz; F2 (L2) min = 1135 Hz) and for second vowel (F1 (L2) max = 730 Hz; F1 (L2) min = 326 Hz; F2 (L2) max = 2837 Hz; F2 (L2) min = 960 Hz).

Additionally, the duration of the vowel “e” in the diphthong was measured at 0.759 seconds, as the spectrogram in Figure 3 demonstrates its longer realization.

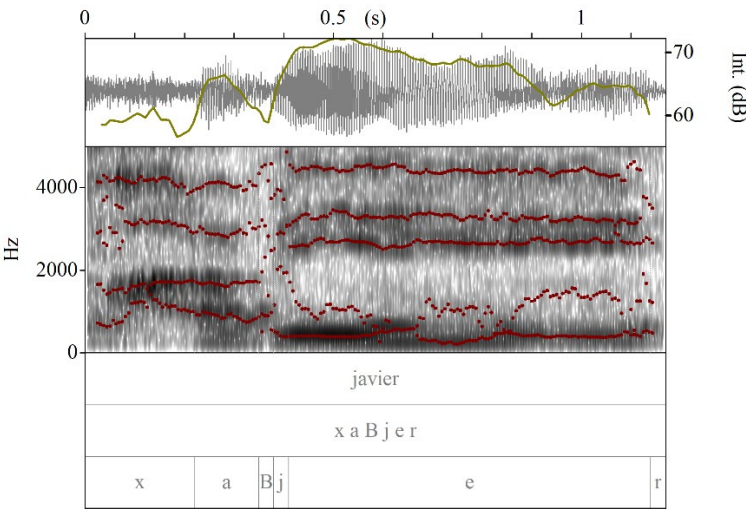


Figure 3. Nominal syntagma “Javier” acoustic realization⁷.

As posited by Bradlow (1995) and Quilis & Esgueva (1983), the F1 and F2 of standard Spanish vowels, which define the manner and place of articulation, are characterised by specific boundaries measured in Hz. The reference boundaries for these formants, as suggested by Bradlow, are as follows:

i: F1: 150–300 Hz; F2: 2100–2400 Hz; e: F1: 450–600 Hz; F2: 1800–2100 Hz.

By applying the approach proposed by Cao, Font-Rotchés, and Rius-Escudé (2024), we observed statistically significant differences in the precision of both vowels ($p = 2.5$ for the first vowel and $p = 16.9$ for the second vowel). This suggests that female native English speakers learning Spanish articulate both vowels in a manner that significantly deviates from the target language, which potentially confuses the model. Furthermore, the duration of the second vowel in the toneme (the last accented syllable) and the overlap between the F1 and F2 ranges appear to contribute to the model's confusion.

These findings suggest that Whisper-large-v3 is capable of identifying speech segments produced differently by non-native speakers compared to native speakers, highlighting areas for improvement. However, the model does not appear to consider all relevant acoustic features for accurate speech recognition. For instance, it failed to account for the duration of the accented vowel in the toneme, which it could not correctly recognise.

Both models, SeamlessM4T and Whisper-large-v3, also seem to struggle with distinguishing voiceless consonants. As shown in Figure 4 (below), the substituted segments exhibit descendant movement in the frequency contour due to both consonants being voiceless. In this context, Whisper-large-v3 predicts “plazos” instead of “platos”, substituting the voiceless alveolar consonant “t” for “z”, which shares the same place of articulation (alveoli). The Fraction of Locally Unvoiced Frames (FLUF) for “t” is 73.33%. Similarly, in the case of “brazos” instead of “platos”, the consonant “l” is replaced with “r”, both of which share the same place of articulation. Furthermore, “p” (FLUF: 100%) and “b” seem to be confused due to their shared manner and place of articulation.

⁷ This spectrogram is automatically generated using Elvira García (2022) original script.

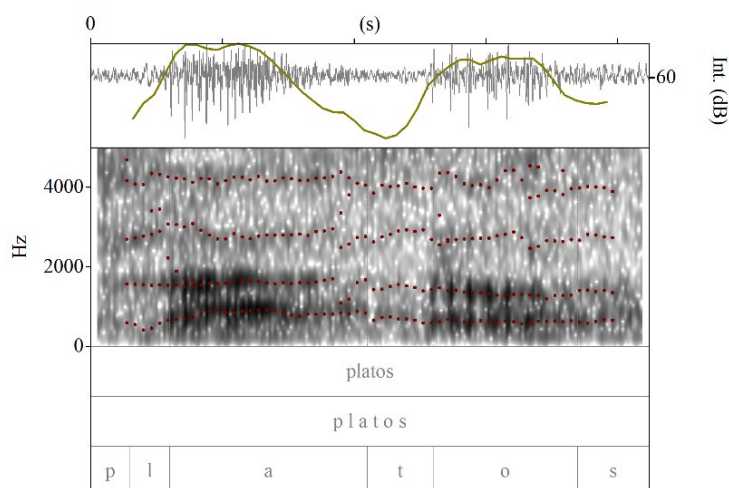


Figure 4. Nominal syntagma “platos” acoustic realization.

This analysis highlights specific challenges in the acoustic realization of vowels and consonants that likely contribute to the observed errors in the ASR model.

Regarding the remaining errors produced by the models, a phonetic analysis is not applicable since the models switch automatically to other languages, producing outputs that are entirely unrelated to the reference utterance. For instance, we present some of these outputs for German L1 speakers as follows:

Reference: sí me gusta viajar solo

Whisper-large-v3: סי מגוסטב יחס עולו

Reference: sí a quién no le gusta viajar

Whisper-large-v3: да как я не могу съехать

These results highlight a limitation of our study, as we are unable to determine the specific causes of these errors. Importantly, these outputs do not seem to be influenced by the L1 of our non-native Spanish speakers, suggesting that the errors are likely unrelated to cross-linguistic interference. Instead, the cause may lie in the model's architecture or the characteristics of its training dataset. As previously mentioned, both SeamlessM4T and Whisper-large-v3 are trained on multilingual data. However, the models struggle to recognise non-native acoustic features correctly, likely because their training datasets did not adequately or equitably represent these features.

6. Conclusion

The study presented in this work aims to contribute to the field of pronunciation assessment of L2 Spanish speakers via an Automatic Speech Recognition framework. In summary, we emphasise the following key contributions, as outlined in Section 1, which, to the best of our knowledge, offer novel insights into the application of ASR for this topic:

- (1) Based on the evaluation of four state-of-the-art ASR models on three distinct datasets of non-native Spanish speakers with varying L1 backgrounds (English, Russian, and German), the Whisper-large-v3 model demonstrated the best performance in the speech recognition task for non-native Spanish speakers. It achieved the lowest WER and CER metrics for both male and female speakers. SeamlessM4T ranked as the second-best performer, showing results that closely followed those of Whisper-large-v3. These

findings provide new insights into ASR performance for Spanish variants, differing from the results reported by San Martín et al. (2024). In contrast, Wav2Vec recorded the highest error rates across all metrics.

- (2) Our study also revealed a notable degree of gender bias. The ASR models consistently performed better on female speech compared to male speech. This disparity highlights the need for more gender-balanced datasets during the training phase to enhance the models' overall performance.
- (3) A qualitative analysis of the ASR model outputs revealed a consistent pattern of substitution errors across the four models and the L2 Spanish speaker datasets. Additionally, we identified a less common error: hypercorrection. This type of error provides valuable insights into L2 Spanish pronunciation assessment by highlighting segments within the speech contour that may require improvement.
- (4) Our phonetic categorization of errors suggests that ASR models face difficulties when vowel formant boundaries of L2 Spanish speakers exceed those of standard Spanish vowels or when voiceless consonants are affected by phonetic assimilation processes. This result is confirmed by precision measurements conducted using the framework proposed by Cao, Font-Rotchés, and Rius-Escudé (2024). Nonetheless, we observed that not all acoustic fundamental features are taken into account by the ASR model, such as duration, in the speech recognition process. Therefore, while ASR models may suggest outputs related to pronunciation assessment for L2 Spanish speakers, these outputs cannot be fully validated without a detailed phonetic analysis.

Regarding the limitations of our study, we used different ASR models, which exhibited varying levels of performance (e.g., Whisper-large-v3 outperformed Wav2Vec). However, we cannot explain why certain models performed better or worse in specific contexts, as these ASR models are *black boxes*, meaning we do not have access to their architecture or internal mechanisms. This limitation means that the findings are specific to the models tested and may not be generalizable to other ASR systems or models not included in the study. Additionally, our study focuses on non-native Spanish speakers with L1 backgrounds in English, Russian, and German, without exploring how the results might vary for speakers with other L1 languages. As such, the diversity of L2 Spanish speakers is not fully addressed, which could limit the generalizability of the findings across all L2 Spanish learners. Furthermore, our study acknowledges that not all relevant acoustic features, particularly aspects like duration, are considered by the ASR models. As a result, the outputs from these models may not capture all the nuances of L2 Spanish pronunciation. This suggests that current ASR models may not provide a comprehensive assessment of pronunciation, especially regarding features such as duration, which are crucial for an accurate evaluation.

For future research, we aim to contribute to the fine-tuning of existing ASR models by creating extensive datasets featuring a larger number of samples from L2 Spanish speakers. This initiative aims to improve the accuracy of ASR models in recognizing L2 Spanish speech and mitigate the observed biases and error patterns. Furthermore, additional investigation is necessary to explore the causes and nature of the identified ASR errors. Future studies could focus on analyzing the training data to identify gaps or biases, thereby enhancing our understanding of language-switching behavior, and examining how the architecture handles multilingual input. Addressing these issues is crucial for enhancing the reliability and accuracy of automatic speech recognition systems, particularly for non-native or L2 speakers.

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Authorship Contribution Statement

Albina Sarymsakova: Conceptualization, Data curation, Resources, Software, Formal analysis, Research, Funding acquisition, Investigation, Methodology, Visualization, Writing (original draft), Writing (review and editing).

Patricia Martín-Rodilla: Conceptualization, Data management, Formal analysis, Writing (review and editing), Funding acquisition, Investigation, Methodology, Project administration, Supervision, Validation, Writing (review and editing).

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